

Semantic Cloze Ranking as a Lightweight Auxiliary Objective for Strict-Small BabyLM Pretraining

Anonymous ACL submission

Abstract

This paper describes a compact masked language model submitted to the BabyLM 2026 Strict-Small track. The system uses a small BERT-style encoder trained on the official Strict-Small corpus with a multi-task objective. In addition to masked language modeling, training includes replaced-token detection, connective prediction, definiteness prediction, collocation discrimination, grammar minimal-pair ranking, and a small semantic cloze ranking task. The semantic cloze task presents a masked prompt with a plausible and implausible single-token completion and trains the MLM head to rank the plausible token higher. The final model has 11.2M parameters and remains compatible with Hugging Face `AutoModelForMaskedLM`. Official evaluation shows that the selected final checkpoint improves EWoK and reading-oriented metrics over a local MLM baseline, while lowering BLiMP and entity tracking.

1 Introduction

The BabyLM Challenge asks whether language models can learn useful linguistic and semantic generalizations from developmentally plausible data budgets (Warstadt et al., 2023; Choshen et al., 2026). This submission focuses on the Strict-Small setting, where the data budget is 10M words. Instead of increasing model size or using external teacher models, the system keeps a compact encoder architecture and changes the training signal.

The motivating hypothesis is simple: a small masked language model may benefit from auxiliary objectives that make lexical, relational, and plausibility contrasts explicit while still using the same MLM prediction head at submission time. The main added objective is semantic cloze ranking. Each example contains a prompt with one [MASK] token, a plausible completion, and an implausible completion. The model is trained to assign higher MLM log-probability to the plausible completion.

Task	Sampling weight
Masked language modeling	0.600
Replaced-token detection	0.100
Connective prediction	0.075
Definiteness prediction	0.075
Collocation discrimination	0.050
Grammar minimal-pair ranking	0.050
Semantic cloze ranking	0.050

Table 1: Training task mixture for the selected system.

This gives the model a direct plausibility-ranking signal without adding a separate classifier head.

2 Model

The submitted model is a compact BERT-style masked language model (Devlin et al., 2019). It uses the `bert-base-uncased` WordPiece tokenizer and a randomly initialized encoder with 4 transformer layers, hidden size 256, 4 attention heads, intermediate size 1024, and maximum position embedding length 512. The maximum training sequence length is 128. The exported model has 11,201,338 parameters and is uploaded publicly at:

<https://huggingface.co/alonsopg/babylm-2026-semantic-cloze-strict-small>

3 Training Objectives

Training uses a weighted multi-task mixture. The primary task remains standard masked language modeling. Auxiliary tasks are generated from the BabyLM Strict-Small training data or from small hand-written semantic templates. No official evaluation items are used to construct the auxiliary data.

The semantic cloze objective uses prompts such as “A shovel is used for [MASK].” with a plausible single-token completion and an implausible alternative. Let s_g and s_b be the MLM log-probabilities of the good and bad completions at the masked position. The ranking loss is:

$$\mathcal{L}_{cloze} = \log(1 + \exp(-(s_g - s_b))). \quad (1)$$

This objective keeps inference unchanged: the final model is still a normal masked language model.

4 Optimization

Training used AdamW with maximum learning rate 5×10^{-4} , weight decay 0.01, gradient clipping at 1.0, batch size 96 sequences, and linear warmup for 6% of training steps followed by decay. Evaluation on a held-out validation split was performed every 500 steps. Early stopping monitored validation MLM loss with patience 2, minimum improvement 0.01, and activation after step 9000. The selected final run stopped at step 9500; the best validation checkpoint occurred at step 5000.

5 Results

Table 2 compares the selected system against a local MLM baseline and the best-validation semantic-cloze checkpoint using the official fast-evaluation helper. The final semantic-cloze checkpoint was selected because it produced the strongest EWoK and reading scores among the retained systems.

The full official evaluation artifact for the selected public model contains predictions for BLiMP, BLiMP Supplement, EWoK, entity tracking, COMPS, reading, SuperGLUE-style finetuning tasks, AoA, and checkpoint-level fast evaluation. Observed full-evaluation scores are shown in Table 3.

6 Discussion

The results suggest that semantic cloze ranking can modestly improve conceptual plausibility and reading-oriented metrics for a compact MLM. However, the trade-off is visible: BLiMP is lower than the local MLM baseline, and entity tracking is sensitive to checkpoint choice. The best-validation semantic-cloze checkpoint improved entity tracking substantially, while the final checkpoint improved EWoK and reading but reduced entity tracking. This supports the view that small-model pre-training objectives can shift evaluation behavior unevenly across linguistic, semantic, and discourse-level probes.

7 Submission Caveat

The public model repository contains all required BabyLM checkpoint revisions, and the collated submission artifact includes AoA plus checkpoint fast-evaluation results for all 19 required Strict-Small

revisions. The selected main model remains the stronger original final checkpoint. The `ckpt_*` trajectory used for AoA and checkpoint fast evaluation comes from an isolated rerun with the same configuration, because the original run did not export Hugging Face revisions during training. Thus, the artifact is complete, but the checkpoint trajectory and selected main checkpoint do not come from the exact same local training run.

8 Limitations

This is a single-seed experiment with a small number of auxiliary objective variants. The semantic cloze templates are intentionally simple and may not cover the full diversity of conceptual knowledge probed by EWoK. The current results should therefore be read as evidence that the objective is promising, not as a definitive statement about the best auxiliary-task mixture.

9 Ethics Statement

The submitted model is trained on the official BabyLM Strict-Small data and small auxiliary examples derived without external personal data. The model is small and intended for research evaluation. No claim is made that it is safe for deployment without additional testing.

References

- Leshem Choshen, Ryan Cotterell, Mustafa Omer Gul, Jaap Jumelet, Tal Linzen, Aaron Mueller, Suchir Salhan, Raj Sanjay Shah, Alex Warstadt, and Ethan Gotlieb Wilcox. 2026. [BabyLM turns 4 and goes multilingual: Call for papers for the 2026 BabyLM workshop](#).
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. [BERT: Pre-training of deep bidirectional transformers for language understanding](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.
- Alex Warstadt, Aaron Mueller, Leshem Choshen, Ethan Wilcox, Chengxu Zhuang, Juan Ciro, Rafael Mosquera, Bhargavi Paranjabe, Adina Williams, Tal Linzen, and Ryan Cotterell. 2023. Findings of the BabyLM challenge: Sample-efficient pretraining on developmentally plausible corpora. In *Proceedings of the BabyLM Challenge at the 27th Conference on Computational Natural Language Learning*, pages 1–34, Singapore. Association for Computational Linguistics.

System	BLiMP	Supplement	EWoK	Entity	Eye	SPR
MLM baseline	64.33	54.00	50.55	15.56	2.47	2.42
Semantic cloze best validation	54.10	57.60	50.73	36.61	7.56	3.11
Semantic cloze final	55.26	54.40	52.00	27.44	8.18	3.71

Table 2: Official fast-evaluation scores for local comparison systems.

Evaluation section	Score
BLiMP filtered accuracy	55.35
BLiMP supplement accuracy	53.29
EWoK accuracy	50.45
Entity tracking accuracy	27.55
COMPS accuracy	50.63
Reading eye-tracking score	8.18
Reading SPR score	3.71

Table 3: Observed full official evaluation scores for the selected submitted model.

Task	Accuracy	F1	MCC
BoolQ	0.6679	0.7755	0.2061
MNLI	0.4269	–	–
MRPC	0.6912	0.8131	0.1300
MultiRC	0.5858	0.2132	0.0859
QQP	0.7044	0.5816	0.3554
RTE	0.5468	0.5191	0.0910
WSC	0.5769	0.1538	-0.0381

Table 4: Observed finetuning scores from the official evaluation pipeline.